

Lesson 11

Managing Service and Data Availability

Topic 1A

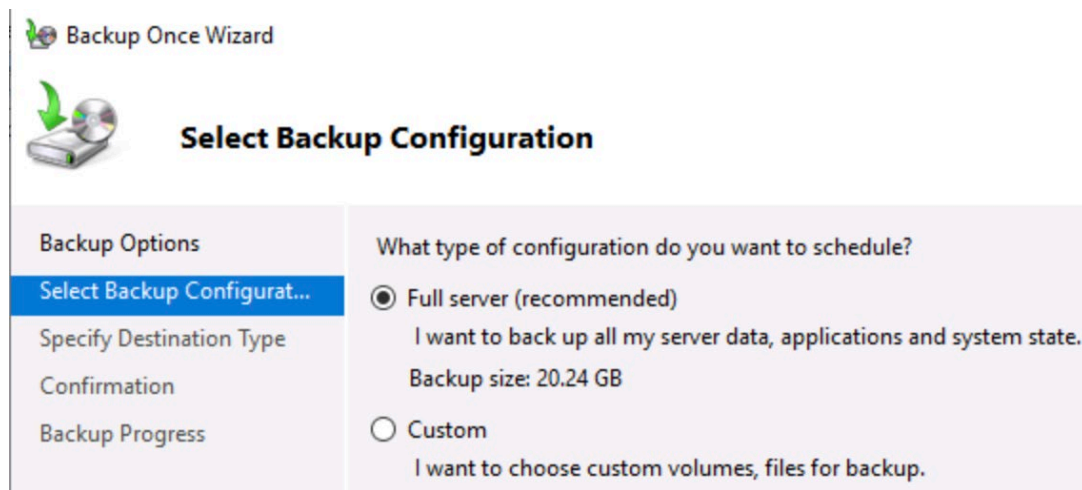
Manage Data Backup and Restore

Exam Objectives Covered

- 3.7 Explain the importance of backups and restores

Managing Backups and Restores

- Full
- Synthetic full
- Incremental
- Differential
- Continuous Data Protection
- Near CDP
- Snapshot
- Archive
- Open file
- System-state versus file-level



Full server backup on Windows Server. (Screenshot courtesy of Microsoft.)

Backup Frequency

- Full
- Full+incremental
- Full+differential
- Backup schemes

Backup Media Types

- Backup media types
- Print backups

Media Rotation

- First-in, first-out (FIFO)
- Grandfather-father-son
- Tower of Hanoi

Backup Validation

- Do trial restores to VMs periodically
- Log files help diagnose issues

Restoring Data

 Recovery Wizard



Getting Started

Getting Started

Specify Location Type

Select Backup Location

Select Backup Date

Select Recovery Type

Select Items to Recover

Specify Recovery Options

You can use this wizard to recover files, applications, volumes, or the system state from a backup that was created earlier.

Where is the backup stored that you want to use for the recovery?

☐ This server (WIN-MSITE54SFL9)

☒ A backup stored on another location

To continue, click Next.

Full server backup on Windows Server. (Screenshot courtesy of Microsoft.)

- Media inventory before restoration

Restore Methods

- Overwrite
- Side-by-side
- Alternate location

Review Activity: Data Backup and Restoration



Topic 11B

Manage High Availability

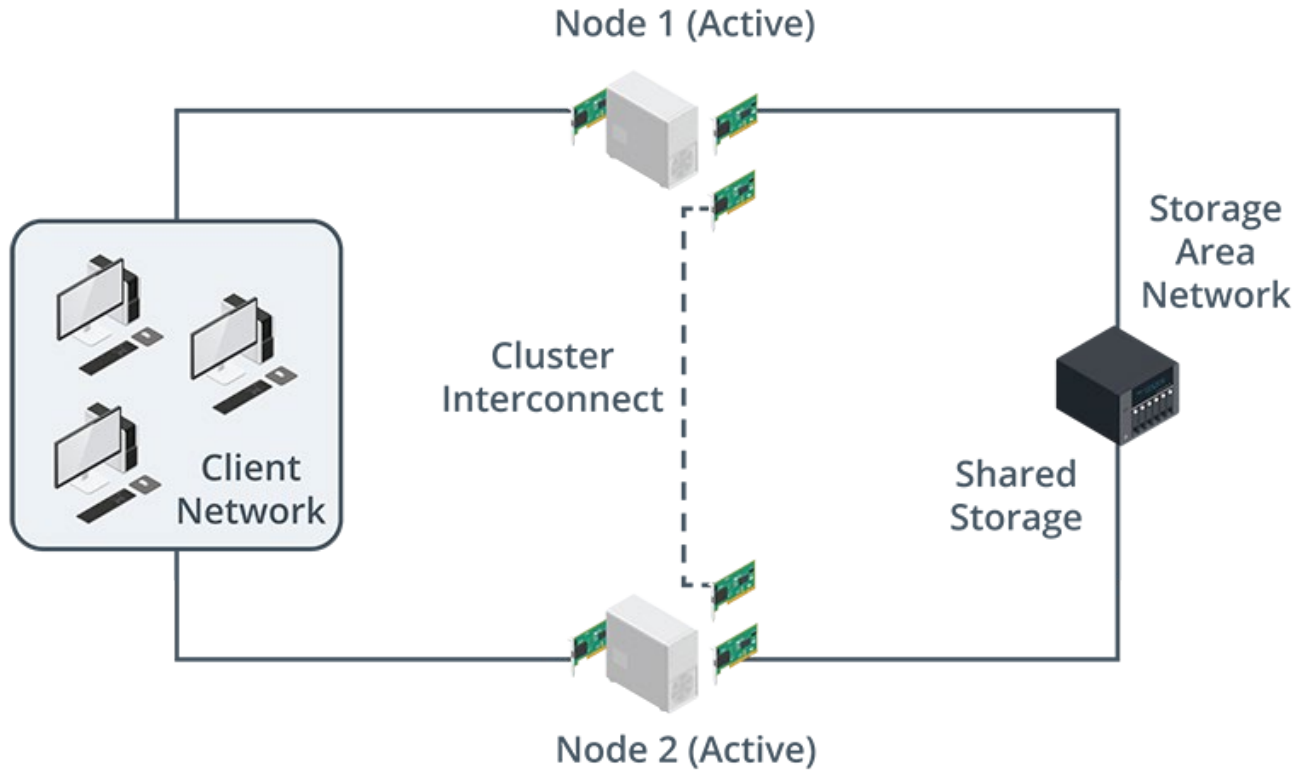
Exam Objectives Covered

- 2.4 Explain the key concepts of high availability for servers

High Availability

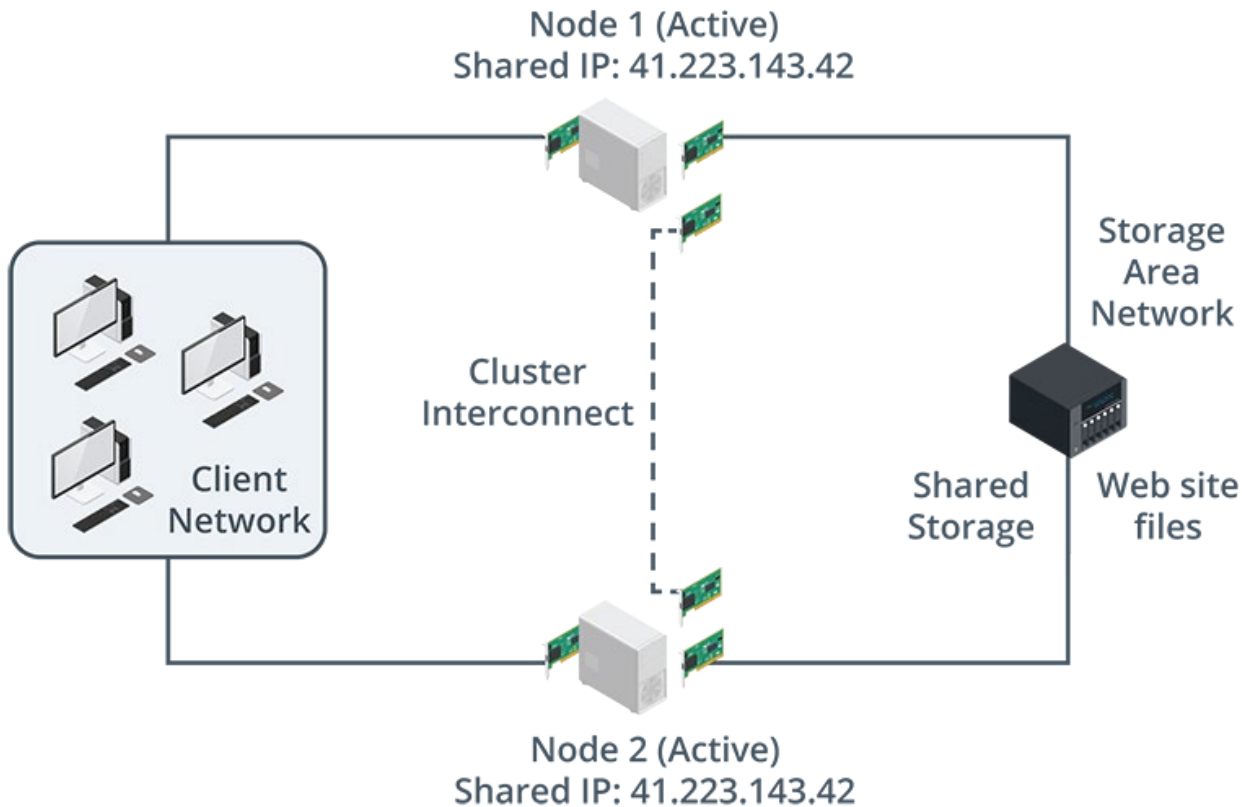
- System of Nines
 - 99.999%
 - 99.99%
 - 99.9%
 - 99%

Server Clusters



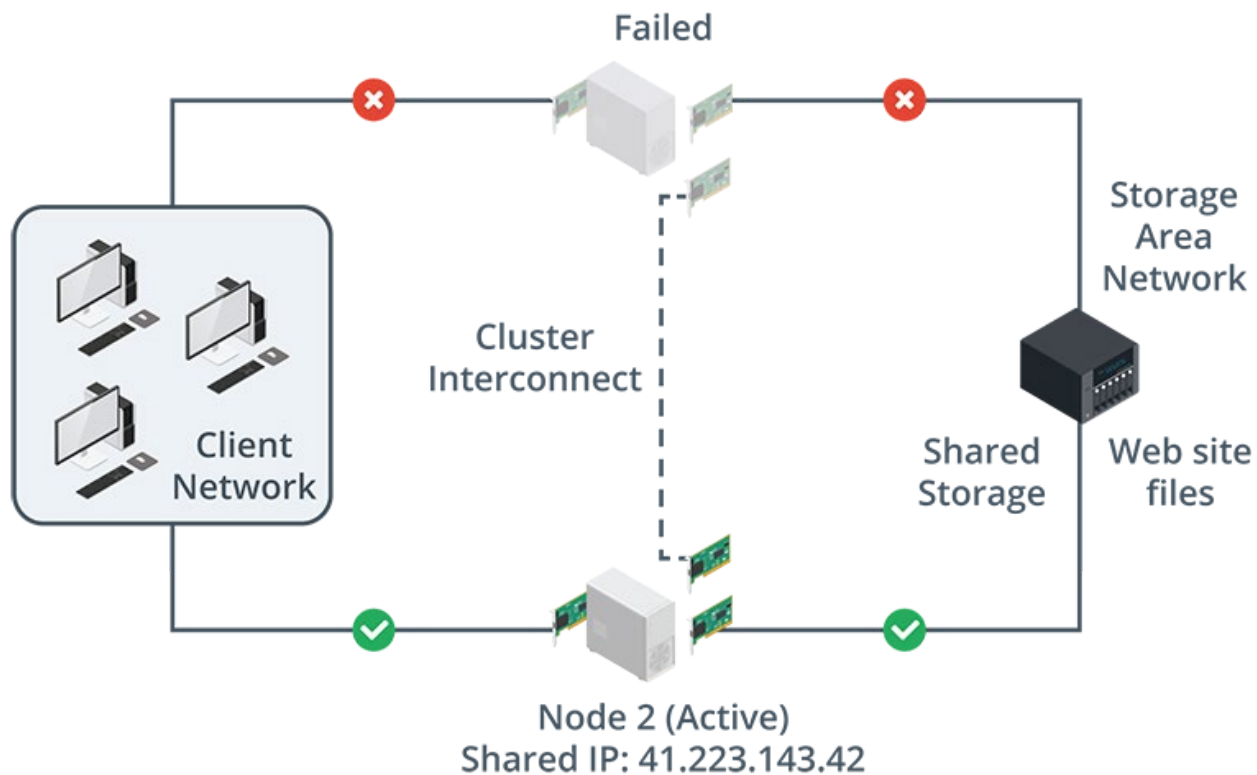
A server cluster. (Images © 123RF.com)

Server Clusters (slide 2)



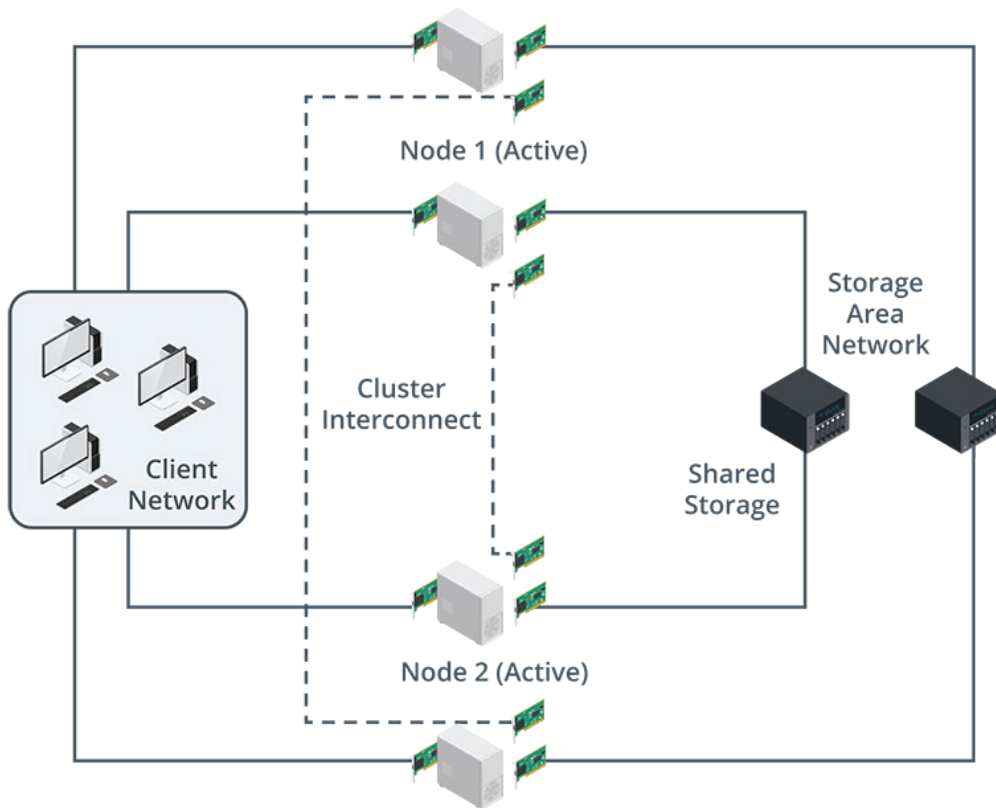
Standard functionality with two web servers sharing the same IP address and web site files. (Images © 123RF.com)

Server Clusters (slide 3)



One server has failed, but the site remains accessible. (Images © 123RF.com)

Avoiding Single Points of Failure

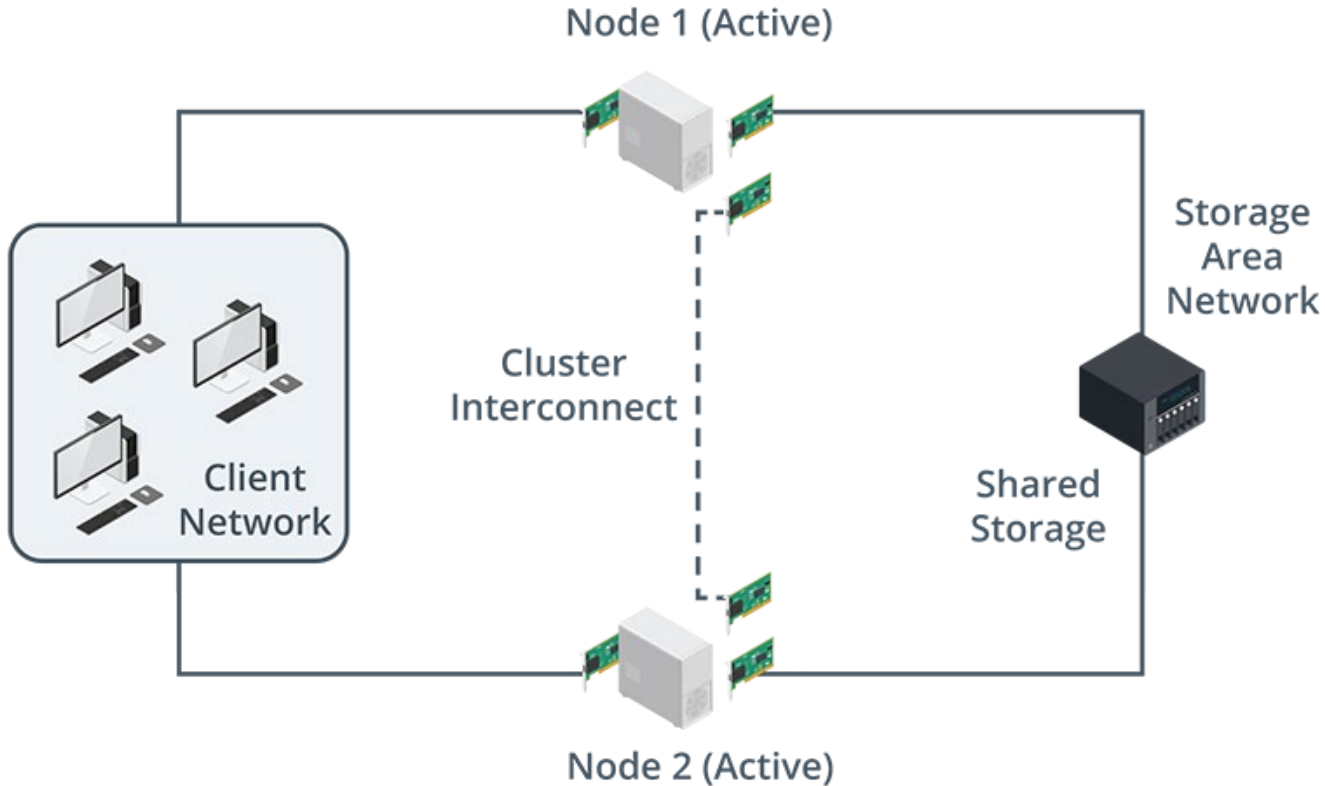


Server cluster with several redundancies. (Images © 123RF.com)

Network Redundancy

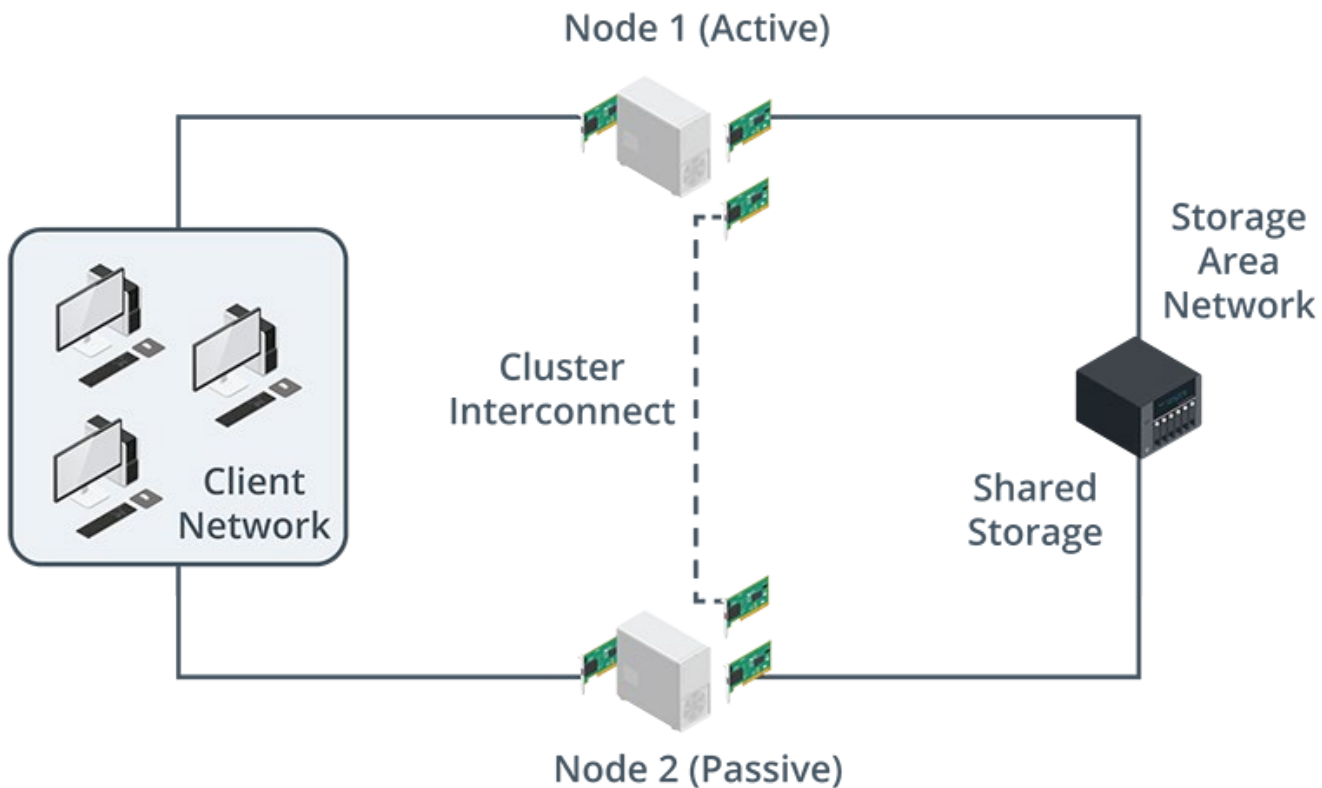
- Cluster heartbeat

Types of Server Clusters



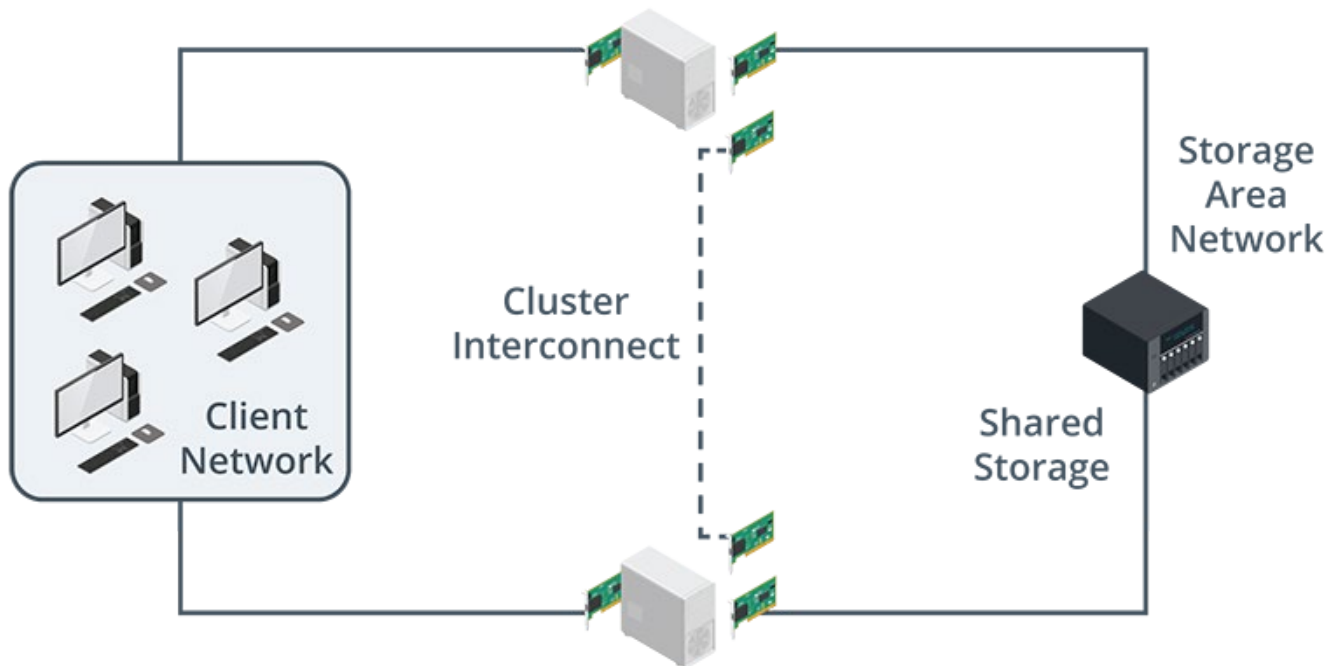
An active-active cluster and load balancer distribute the workload. (Images © 123RF.com)

Types of Server Clusters (slide 2)



An active-passive cluster. (Images © 123RF.com)

Patching Failover Clusters



Node 2 (Passive) is patched and then made active; Node 1 is made passive and then patched. (Images © 123RF.com)

Fault Tolerance

- Redundancy
 - Automated
 - Manual
 - Component

Load Balancing

- Hardware
- Software
- Schedules

Nic Teaming

NIC Teaming ✕

New team

Team name:

DevServer

Member adapters:

In Team	Adapter	Speed	State	Reason
☒	Ethernet	10 Gbps		
☒	Ethernet 2	10 Gbps		

⌵ Additional properties

OK Cancel

Configuring NIC Teaming in Windows Server. (Screenshot courtesy of Microsoft.)

Review Activity: High Availability



Review Activity

Topic 11C

Manage Disaster Recovery

Exam Objectives Covered

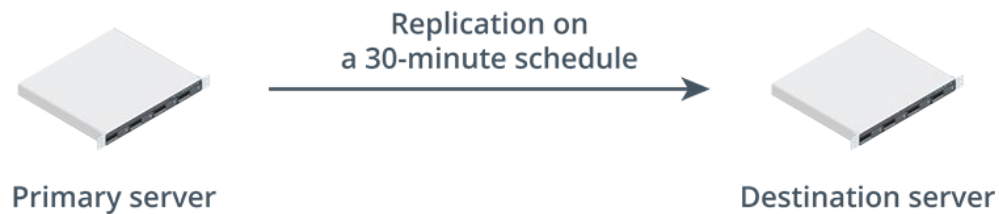
- 3.8 Explain the importance of disaster recovery

Sites

- Hot
- Warm
- Cold
- Geographically separate
- Cloud

Data Replication

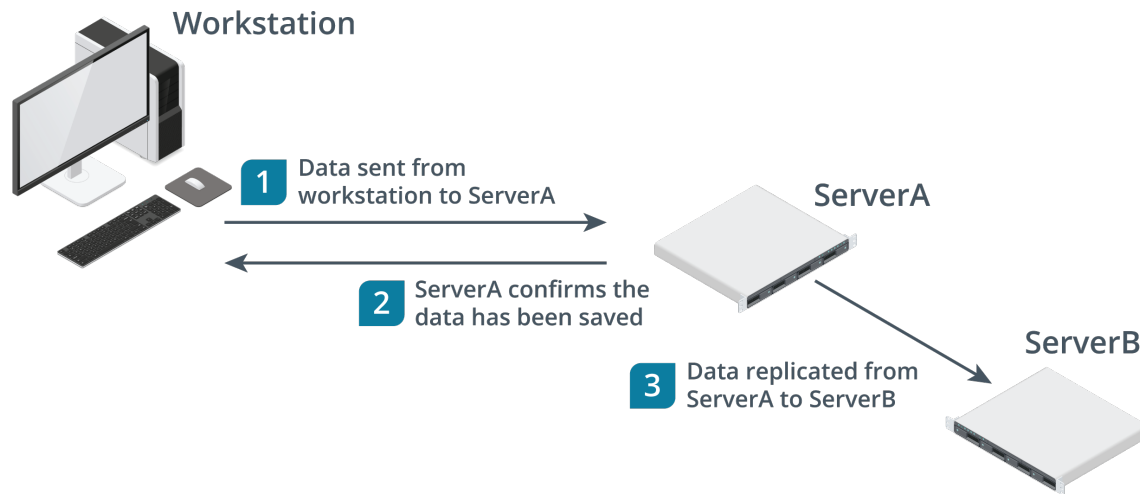
- Examples
- Constant
- Periodic



Data replication on a regular timetable. (Images © 123RF.com)

Data Replication (slide 2)

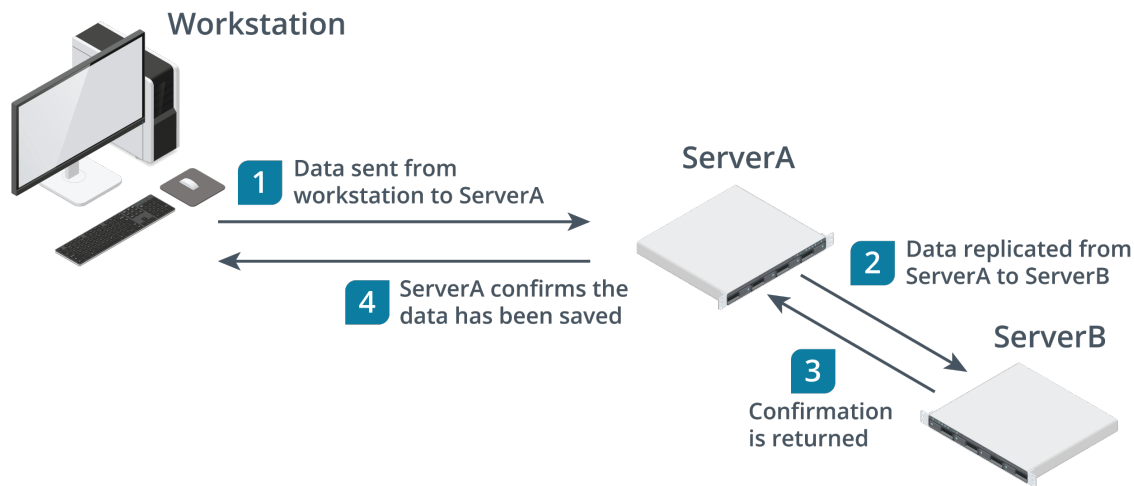
- Asynchronous replication



The workstation is updated that the data was saved, even though the data has not yet been replicated to Server B. (Images © 123RF.com)

Data Replication (slide 3)

- Synchronous replication



The workstation is not updated that the data was saved until the data has been replicated to Server B. (Images © 123RF.com)

Data Replication (slide 4)

- Application consistent
- File locking
- Mirroring
- BiDi

Data Recovery

- Disaster recovery
 - Planning
 - Testing
- Testing types

Guidelines for Managing Disaster Recovery (DR)

- Sites
 - Hot
 - Warm
 - Cold
- Replication
- Test DR plans regularly

Review Activity: Disaster Recovery



Review Activity

Lesson 11

Summary

